

rPower

Write a **recursive** function that computes the power of a number `num`. The power `p` may be any integer value. Write the recursive function in two versions. The function `rPower1()` computes and returns the result. The function `rPower2()` computes and returns the result through the parameter `result` using call by reference. The function prototypes are given as follows:

```
float rPower1(float num, int p);  
void rPower2(float num, int p, float *result);
```

A sample program template is given below to test the functions:

```
#include <stdio.h>  
float rPower1(float num, int p);  
void rPower2(float num, int p, float *result);  
int main()  
{  
    int power;  
    float number, result;  
  
    printf("Enter the number and power: \n");  
    scanf("%f %d", &number, &power);  
    printf("rPower1(): %.2f\n", rPower1(number, power));  
    rPower2(number, power, &result);  
    printf("rPower2(): %.2f\n", result);  
    return 0;  
}  
float rPower1(float num, int p)  
{  
    /* Write your code here */  
}  
void rPower2(float num, int p, float *result)  
{  
    /* Write your code here */  
}
```

Some sample input and output sessions are given below:

- (1) Test Case 1:
Enter the number and power:
2 3
rPower1(): 8.00
rPower2(): 8.00
- (2) Test Case 2:
Enter the number and power:
2 -4
rPower1(): 0.06
rPower2(): 0.06
- (3) Test Case 3:
Enter the number and power:
2 0
rPower1(): 1.00
rPower2(): 1.00